

Course Syllabus

Course Title **EM 649 - MANUFACTURING PROCESS PLANNING**

The website for the class will be located at www.emuonline.edu and will be accessible on .

Course Description **Graduate Catalog Description:** The design of the manufacturing process as required by product design specifications. Selection of processes, tooling, work-holding, gauging, routing, and material handling, as well as the development of the human interface.

Terminal Objectives Upon completion of all this courses in Design and Manufacturing Track, students will be able to analyze and evaluate different manufacturing processes, select the proper sequence of processes to achieve the desired output, as well as feedback to the design process as part of concurrent engineering.

Specifically, students will be able to:

- Apply major manufacturing processes where appropriate
- Analyze and evaluate any given manufacturing processes
- Select a proper sequence of processes to produce the part for a given design of a product
- Feedback to the design process as part of concurrent engineering

Course Objectives At the completion of this course, the student should be able to:
A. Describe traditional mass production and modern lean manufacturing systems and tools for production planning and operations management.
B. Analyze product characteristics for process design.
C. Identify production processes needed for product manufacture.
D. Develop production process documentation needed for product manufacture.

Text Book **Required :**
Standard, C., & Davis, D. (1999). Running today's factory. Dearborn, MI: Society of Manufacturing Engineers. (ISBN 0-87263-513-9)
<http://www.sme.org> or 1.800.733.4763
Optional:
(1993). Computerized Manufacturing Process Planning Systems; 1 edition. London, England: Springer. (ISBN 10:041241300). *This book available at a very low cost at Amazon.com and free online at googlebooks.*
Googlebooks: [Computerized manufacturing Process Planning](#)
Amazon: [Buy it Cheap "Computerized Manufacturing Process Planning"](#)

Course Outline and Schedule **May be subject to change.**
All Chapters are from the Text Book

Unit 1.
Reading and review: Chapters 1, 2. Manufacturing issues and challenges.
Introduction.

Unit 2. (Two weeks)

Reading and review: Chapters 3, 4, 5. Modern manufacturing system basics.

Process planning project

Unit 3. (Two weeks)

Reading and review: Chapters 6, 7, 8. Modern manufacturing system strategies.

Unit 4. (Two weeks)

Reading and review: Chapters 9, 10. Production control and quality.

Process planning project

Unit 5. (Two weeks)

Reading and review: Chapters 11, 12. Setup, lot size, and metrics.

Process planning project

Unit 6. (Two weeks)

Reading and review: Chapter 13. Management commitment.

Process planning project

Unit 7. (Two weeks)

Process planning project

Unit 8.

Final exam

Process planning project

Peer evaluation

General Policies

A. Students are encouraged to discuss problems related to the class with the professor via telephone or email.

B. Academic Dishonesty: The policy of the School of Engineering Technology is that any student found to have engaged in any activity constituting academic dishonesty will receive an "E" for the course in which the activity

occurred. This policy relates to all forms of work associated with the course requirements (tests, lab work,

papers and other assignments). Academic dishonesty includes: cheating on quizzes and exams, copying, or

taking credit for another's work, plagiarism, etc. Graduate students enrolled at EMU should be aware of academic dishonesty policies and consequences, and examples of what constitutes academic dishonesty, from

EMU Academic Affairs, EMU Student Judicial Services, EMU Graduate School, and the School of Engineering Technology.

My managerial approach regarding minor occurrences of academic dishonesty is to inform the involved student(s) about the problem, give a warning to the student(s) of the possible consequences, and share an

example of the problem with the rest of the class, as a learning experience for the whole class. The consequences may be more severe (such as failure of the assignment or the entire course), and without a

warning, for second and/or major occurrences of academic dishonesty.

C. Writing Style and Format: It is the policy of the School of Engineering Technology (SET) that students should follow the APA (American Psychological Association) technical writing style and format guidelines for

papers submitted in all SET classes. Some professors will emphasize use of APA guidelines more than others.

It is recommended that you get a copy of the 6th edition of the APA Publication Manual as soon as possible, as

you will use it throughout your degree. You should especially be familiar with traditional and electronic

referencing guidelines for quoting, paraphrasing, and referencing information that you use, to help avoid Academic Dishonesty problems, as mentioned above. See <http://www.apa.org/journals/acorner.html#pubmanual> for more specific information and examples.

D. Incomplete Final Grades (I): will be given only in accordance with EMU's guidelines, which are stated in the 2007-2009 Graduate Catalog at <http://catalog.emich.edu/index.php?catoid=2>, as follows: I (Incomplete) Grades - An I grade may be awarded when illness or other unavoidable extenuating circumstances prevent completion of a course, provided that academic performance for the completed portion of the course merits a grade of B or better.

E. Student and Exchange Visitor Information Service (SEVIS): For students who hold an F or J visa, to avoid being out-of-status, it is your responsibility to report within 10 days to the Office of International Students (OIS), King Hall, any change in name, current address of residence, academic status (full or part-time enrollment), date of program completion, program/major change, level change (undergraduate to graduate), change in funding source (employment or graduate assistant position), probation or disciplinary action leading to suspension due to conviction of a crime. Effective January 1, 2003, EMU has 21 days from the date of the reported event to inform the U. S. Government -- Department of Homeland Security, Bureau of Border and Transportation Security and the Bureau of Citizenship and Immigration Services -- SEVIS database (functions of the formerly known as Immigration and Naturalization Service are a part of two new offices effective March 1, 2003). OIS enters the data into a real-time database and new documents or I-20s are then issued to you. Remember that dropping or withdrawing from course(s) may put you below full-time enrollment. Before a drop/withdrawal change is made that reduces enrollment below full-time status, you must receive permission from the OIS. If you plan to transfer to another university, this process must begin before the end of your last semester at EMU, contact the Office of International Students. Noncompliance could result in actions leading up to a fine, arrest, or deportation per the U.S. Government.

Participation Guidelines

Here are some guidelines for making things go smoothly.

A. Access to Course.

Our course consists of 8 units, which are scheduled from Mondays through Sundays. Most units will be scheduled across two weeks, but Unit 1 is 12 instead of 14 days long (due to a Wednesday, rather than Monday start date) and Unit 8 is scheduled across just one week (due to Finals week). Most assignments for each unit will be due at the end of the unit, on Sunday nights, by 11:59 PM ET, including the last unit, which ends on **SUNDAY 12/18/2011**. In some cases, some "preliminary" assignments will be due mid-way through the unit. I will usually open access to each unit early, on Fridays, so that you can get an early start working on it if you like; therefore, you will have three weekends available for working on most of the units. All completed units will remain open for review, until **SUNDAY 12/18/2011**.

B. Participation.

In online courses, students seek interaction with other students and with the instructor. Since there are no required face-to-face meetings, this interaction becomes very important to effective learning online. As such, good class participation is essential and will account for a significant part of your grade. [Plan to contribute to every discussion topic with a "quality" contribution -- i.e. a response that expresses clear thinking, and that is relevant to the discussion. This usually involves more than just "I agree."].

C. Online Conduct.

I believe that discussion of the following is probably not very necessary. Nevertheless, for the sake of clarity, please realize that the University and I cannot and will not be responsible for the content of any personal messages that are sent from one student directly to another student using the online email delivery system. It is expected that all students will adhere to accepted codes of ethical, personal, and civil conduct when conversing online, using email, or engaging in any real time online chat sessions. Failure to abide by such codes of conduct and etiquette may result in withdrawal from the course and a failing grade.

D. Email to Classmates.

Students are free to use the "email the class" feature on the course web site. Try to keep the messages sent to the whole class related to course content. Also, when emailing the class, generally it is best NOT to email phone numbers or addresses unless it is being emailed directly to a specific individual or the instructor. This helps maintain privacy.

E. Email to Instructor.

For EM 649 course-related submissions and communications, please send all email directly to me at my personal email account, at gvrsek@emich.edu. Also, in the subject line of your email messages, include the: initials of your full name, course number, and the subject of the message (such as the name of an assignment that is attached to your email); for example, Subject: XYZ649Design1. I can not be on call for your class 24 hours a day, but will be checking in as often as possible. I expect to check EM 649-related email before 12 noon on several days during the week. Due the amount of email I will receive from students, as well as other work and travel commitments, I may not be able to respond to emails or confirm receipt of submitted assignments more than once or twice a week--although I will try to do so more frequently. So, plan accordingly. If you wait until the last minute to do an assignment, and then email a request for assistance with a problem, neither I nor a fellow student may be able to respond quickly enough to help you. Also, it may take a day or two before you receive an emailed confirmation from me that I received your submitted assignment. Typically, the confirmation will read: "Got it, "

F. Email Attachments.

There are so many different word processing programs that email attachments can become a problem in online courses if there is no standard policy set for them. It is preferred that you send any paper that is an attachment to an email in MS Word. Formats that are compatible with MS Word include earlier Word versions, Microsoft Works, Word Perfect 5 or 6, and Rich Text Format are also acceptable. If necessary, you can compose the work in your non-compatible word processor and simply cut and paste it into an email message-- although this may cause a problem if formatting is important. If using MS Project software, please submit your work in MS Project 2002 format. You may also send single or multiple files in .zip compressed format. The file name of your attachment should be similar to the subject line of your email message, including the: initials of your full name, course number, and the name of the attachment with its file type; for example, the file name of an attached assignment would look like: XYZ649Design1.pdf.

G. Grammar, Spelling, Writing Style, and Format.

We all tend to become somewhat lax when posting responses to threaded discussions or when sending email messages. Grammar and spelling is important to me. I am not a fanatic about it, but I would appreciate your composing threaded discussion responses or email messages in a word processing program where you can check spelling, edit, and revise your writing. Then you can cut and paste the work into the threaded discussion or email message. You can do the same for other coursework and thereby help with NOT tying up Internet Service Provider lines. Good grammar, spelling, writing style, and APA format is required for formal papers, per IT Department guidelines.

H. Assignment Delivery and Technical Problems.

In most cases, assignments and threaded discussion responses are to be turned in and completed before or on Sundays, by midnight (11:59 PM ET), before the next unit begins. Assignments turned in late will be penalized 20% per day, rounded up or down to the nearest whole point, and will not be accepted if more than 3 days late, (after Wednesday midnight, 11:59 PM ET). Since you will have more than two weeks and three weekends available for completing your work throughout most of the course, I expect you to get it done on time. Final projects and the final exam (and any other assignments due in our Unit 8 final week) will be due by **SUNDAY 12/18/2011**midnight), and will not be accepted late. Because of the nature of the EMU Online examination administration system, exams taken or submitted late can not be accepted. Let me encourage you to make hard copies of your assignments and communications with me. Computer crashes and disasters are bound to happen. Keep in mind that I do not view ISP problems as an excuse for turning work in late. At the first sign of problems with email, contact your ISP immediately. If your problems are not cleared up within 12 hours, send in the assignment in one of the following ways:

1. Fax it to me at 734.487.8755 (EMU).
2. Overnight express hard copy and on disk.
3. Make arrangements over the phone with me.

Note: It is your responsibility to access the course on another computer in the event of hardware or software problems. This can be done with a friend's computer, or one at Kinko's or the Public library. My guidance from the university is to not consider any excuse for missing assignments due to technical difficulties as legitimate.

I. Personal Involvement and Time.

Students often have the misperception that taking a course online is easier since they don't have to physically go to a classroom. When taking an online course there are some things that become easier for students. For example, it becomes easier NOT to go to class. It also becomes easier to get behind. This medium gives students flexibility in attending class--but remember, while taking a course online may be more convenient, it is not necessarily easier! Be aware of the general guidelines to professors for student work loads: 45 hours of work per credit hour per semester. This is a 3 credit class, so I expect, on average, that students will spend 90 hours of total work time in this course. Students with more graduate-level and online learning experience tend to be more efficient and require less time--some students will require a great deal more time to get their school work completed. Be prepared for the work involved, as it may require 10 or more hours of your time each week.

J. EMU Writing Support

The University Writing Center (115 Halle Library) offers one-to-one writing consulting for both undergraduate and graduate students. Students can make appointments or drop in between the hours of 9 a.m. and 6 p.m. Mondays through Thursdays and from 11 a.m. to 4 p.m. on Fridays. Students should bring a draft of what they're working on and their assignment.

The UWC also offers small group workshops on various topics related to writing (e.g., Reading in College: Tips and Strategies; Incorporating Evidence; Revising Your Writing). Workshops are offered at various times Monday through Friday in the UWC. To register for a workshop, click the "Register" link from the UWC page at <http://www.emich.edu/english/writing-center>.

The Academic Projects Center (116 Halle Library) offers one-to-one consulting for students on writing, research, or technology-related issues. No appointment is required – students can just drop in. The APC is open 11-5 Monday-Thursday. Additional information about the APC can be found at <http://www.emich.edu/apc>. Students visiting the Academic Projects Center should bring with them a draft of what they're working on and their assignment sheet.

Grading Policies EVALUATION.

Course grades will be determined by the percentage of total points attained for various work assignments completed throughout the semester, based upon the following scale:

100 to 93% = A
92.99 to 90% = A-
89.99 to 87% = B+
86.99 to 83% = B
82.99 to 80% = B-
79.99 to 77% = C+
76.99 to 73% = C
72.99 to 70% = C-
(no D grades, based on Grad School requirements)
69.99 and below = E

Final course grades will be based as shown above, but may be subject to adjustment based on the judgment of your professor.

ACTIVITIES

The general format of this class will be through online guidance from the professor, threaded discussion, and submission of assignments which include: reading review assignments and quizzes, peer evaluations of fellow students' work, process planning project work, and a final exam.

A. ONLINE GUIDANCE FROM PROFESSOR. Students will follow the directions given by the professor via the ecollege system and submit completed assignments electronically to the professor.

B. THREADED DISCUSSION. Students will receive points for participation in a threaded discussion on topics posted by the professor that are related to that unit. Primarily, these activities will include:

C. READING REVIEW ASSIGNMENTS AND QUIZZES. Students will read assigned chapters from the course textbook, write and review potential final exam questions over that material, and take quizzes over that material.

D. PEER EVALUATION. Students will read, analyze, and evaluate portions of other students' work. (approx. 180 total points)

E. MAJOR PROJECT. Students will participate in a process planning project as described in the unit.

F. FINAL EXAM. Students will develop a pool of examination questions throughout the semester, based on their readings. A significant portion of the final examination will be made up of questions selected from that pool by the professor.